

Claude Code Prompt

LibreOffice Calc Julian Date Add-In (Java)

Build a free, open-source LibreOffice Calc add-in in Java that converts between calendar dates and Julian representations. Release under the MIT License with cross-platform support (Slackware, Debian, Ubuntu, Windows).

Java constraints (LibreOffice-embedded JVM)

Target **Java 8**. Do not use records, `var`, `java.net.http.HttpClient`, or any Java 9+ API. No third-party dependencies — use only the JDK standard library to avoid classloader conflicts inside LibreOffice's JVM.

Functionality

This add-in is pure computation (no network I/O), so all cell functions can return synchronously. Implement both meanings of "Julian" — clarify with the user if only one is intended, but default to supporting all of the following:

Julian Day Number / Julian Date (astronomical):

`TO_JULIAN_DAY(date)` → Julian Day Number (integer) from a Calc date serial

`TO_JULIAN_DATE(date)` → Julian Date including fractional day/time

`FROM_JULIAN_DAY(jdn)` → Calc date serial from a Julian Day Number

`FROM_JULIAN_DATE(jd)` → Calc date serial (with time) from a Julian Date

`TO_MJD(date)` / `FROM_MJD(mjd)` → Modified Julian Date ($JD - 2400000.5$)

Ordinal / "Julian" day-of-year format (business usage, e.g. YYDDD or YYYYDDD):

`TO_JULIAN_ORDINAL(date, [digits])` → 2025-01-15 → 25015 or 2025015

`FROM_JULIAN_ORDINAL(ordinal)` → Calc serial, auto-detecting YYDDD vs YYYYDDD by length

Julian ↔ Gregorian calendar conversion (proleptic):

`GREGORIAN_TO_JULIAN(date)` and `JULIAN_TO_GREGORIAN(date)`

For historical dates that differ between the two calendars.

Implementation requirements

Use the standard algorithms (Fliegel–Van Flandern or equivalent) for JDN conversion. Handle the Calc date epoch correctly (default 1899-12-30, day 0) and make the null-date offset explicit and documented. Guard against the Gregorian/Julian calendar reform discontinuity where relevant.

LibreOffice add-in structure

Implement as an **Add-In only** UNO component (no dialogs, no toolbar). Include:

- The Java service implementing `com.sun.star.sheet.XAddIn` plus the custom function interface.

- `CalcAddIn.xcu` — **critical: set a `CompatibilityName` for every function** to prevent silent data loss when files are saved/reopened as XLS/XLSX.
- `description.xml`, `META-INF/manifest.xml`, and function metadata (`Function.idl` compiled to a type library, or the equivalent registration).
- A dependency-free build (Ant or plain `javac + jar + zip` into a `.oxt`) — no Maven, no bundled jars.

Deliverables

- Complete source tree ready to build.
- Build script producing the `.oxt` extension.
- `README.md` with per-platform install steps, a function reference table, worked examples for each function, and notes on epoch/offset handling and calendar-reform caveats.
- `LICENSE` (MIT).

Before writing code

Briefly confirm: (1) which Julian meanings to include, and (2) whether ordinal output should default to `YYDDD` or `YYYYDDD`.